Curve-Warp AMMs:

A Geometric Formalization of Weight-Adjusted Liquidity Provision

Temporal Exchange

Abstract

We formalize a class of weighted-geometric AMMs in which trade impact is realized by deforming the pricing curve about a fixed trade point, rather than by sliding reserves along a fixed curve. In log-log coordinates the pricing curve becomes a straight line and the deformation reduces to a rotation about the trade point. We identify the natural intrinsic coordinate—the *rapidity* $\xi = \ln \tan \theta$—and prove a local conservation law: the log-slope is affine in rapidity with coefficient exactly 1 regardless of pool weight. Strikes are characterized by trade-invariant rapidities; the pool's mode rapidity moves under trades by an amount equal to the log of the price impact. Reciprocal-strike pairs (call and put at equal mode-offset rapidity) carry identical option premia independently of weight. Trades that move the mode past a strike's rapidity flip the strike's type (call $\leftrightarrow$ put).

1 Setup

1.1 The weighted-geometric curve

The pricing curve of a pool with state (w, k), $w \in (0, 1)$, $k > 0$, is

$$V(x, y) := x^w \, y^{1-w} = k, \qquad x, y > 0. \tag{1}$$

1.2 Polar coordinates

Parametrize the curve by $\theta \in (0, \pi/2)$ with $\tan\theta = y/x$. Eq. (1) gives

$$x(\theta) = k \, (\tan\theta)^{w-1}, \qquad y(\theta) = k \, (\tan\theta)^{w}. \tag{2}$$

1.3 Slope and option premia

Differentiating (1) along the curve,

$$\frac{dy}{dx} = -\frac{w}{1-w} \cdot \frac{y}{x} = -\frac{w}{1-w} \tan\theta. \tag{3}$$

Write $\sigma(\theta) := \dfrac{w}{1-w} \tan\theta$ for the slope magnitude. The two option premia are read off as side-specific secant ratios:

$$P_C(\theta) := -\frac{dx}{dy} = \frac{\tan\theta_m}{\tan\theta} = \frac{1}{\sigma(\theta)}, \qquad P_P(\theta) := -\frac{dy}{dx} = \frac{\tan\theta}{\tan\theta_m} = \sigma(\theta), \tag{4}$$

where the **mode angle** θ_m is the unique angle with $\sigma = 1$:

$$\tan\theta_m = \frac{1-w}{w}. \tag{5}$$

P_C is the call premium (notional in y, premium in x); P_P is the put premium (notional in x, premium in y, under the X-axis mirror convention). They satisfy the *premium duality*

$$P_C(\theta) \cdot P_P(\theta) \equiv 1, \tag{6}$$

and both equal 1 at the mode—the pool's ATM (premium-neutral) point by construction.

1.4 Log-log linearization and rapidity

Let $u := \ln x$ and $v := \ln y$. Taking logs of (1),

$$w\,u + (1-w)\,v = \ln k. \tag{7}$$

Every pricing curve is a straight line in (u, v) *space*, with slope $-w/(1-w)$, intersecting the diagonal $u = v$ at $(\ln k, \ln k)$—the pool's mode.

Define the **rapidity**

$$\xi := \ln \tan \theta = v - u, \qquad \xi_m := \ln \tan \theta_m = \ln \frac{1-w}{w}. \tag{8}$$

The slope and premia rewrite as

$$\sigma(\xi) = e^{\xi - \xi_m}, \qquad P_C(\xi) = e^{\xi_m - \xi}, \qquad P_P(\xi) = e^{\xi - \xi_m}. \tag{9}$$

1.5 Strikes as trade-invariant rapidities

A strike is a point on the curve fixed in the (x, y) plane and is intrinsically labeled by either its polar angle θ_{strike} or its rapidity $\xi_{\text{strike}} = \ln \tan \theta_{\text{strike}}$. These are coordinate identifiers in the underlying plane and are therefore *invariant* under any trade. By contrast, the strike's offset from the pool's mode—measured either in percent ("+30% from spot") or in any mode-relative angular form—is a *dynamic* reading that shifts whenever the trade moves θ_m.

Remark. The pool's response to a trade—worked out in §5—reduces to a single scalar shift in ξ_m. Since premia at a strike depend only on $\xi_{\text{strike}} - \xi_m$, the entire effect of any trade on every strike on the book is captured by that one scalar.

2 Local invariants

Proposition 1 (Log-slope is affine in rapidity)**.** *For any point on the curve* (1),

$$\ln \sigma(\xi) = \xi - \xi_m. \tag{10}$$

The relationship is affine in ξ *with coefficient identically* 1*, independent of* w *and* k*.*

Proof. From (3) and (8), $\sigma(\xi) = e^{-\xi_m}\, e^{\xi}$. □

Corollary 1 (Premia in rapidity)**.** $\ln P_C(\xi) = -(\xi - \xi_m)$ *and* $\ln P_P(\xi) = \xi - \xi_m$*; in particular* $P_C \cdot P_P = 1$*.*

The pool weight w enters only through the origin ξ_m; the pool scale k does not enter at all. In $(\xi, \ln \sigma)$ coordinates every pool is the same unit-slope line, translated.

3 The conservation law

Theorem 1 (Slope-product invariant)**.** *Fix any point on the curve at rapidity ξ_*. For every $\Delta\xi$,*

$$\sigma(\xi_* + \Delta\xi) \cdot \sigma(\xi_* - \Delta\xi) \;=\; \sigma(\xi_*)^2. \tag{11}$$

Equivalently, $P_C(\xi_* + \Delta\xi)\, P_C(\xi_* - \Delta\xi) = P_C(\xi_*)^2$, and the same with P_P.

Proof. By Proposition 1, $\ln \sigma(\xi_* \pm \Delta\xi) = (\xi_* \pm \Delta\xi) - \xi_m$. Summing yields $2(\xi_* - \xi_m) = 2 \ln \sigma(\xi_*)$. □

The local slope (and each premium) is the geometric mean of its equal-rapidity neighbors. The invariant holds at every point of every pool curve.

Corollary 2 (Hyperbolic structure of slope integrals)**.** *Let $\Sigma_\pm(\Delta\xi) := \int_{\xi_*}^{\xi_* \pm \Delta\xi} \sigma \, d\xi$. Then*

$$\Sigma_+(\Delta\xi) + \Sigma_-(\Delta\xi) = 2\, \sigma(\xi_*)\, \sinh(\Delta\xi), \tag{12}$$

$$\Sigma_+(\Delta\xi) \cdot \Sigma_-(\Delta\xi) = 2\, \sigma(\xi_*)^2\, (\cosh \Delta\xi - 1). \tag{13}$$

Proof. $\Sigma_+ = \sigma(\xi_*)(e^{\Delta\xi} - 1)$, $\Sigma_- = \sigma(\xi_*)(1 - e^{-\Delta\xi})$. Their sum is $\sigma(\xi_*)(e^{\Delta\xi} - e^{-\Delta\xi}) = 2\sigma(\xi_*) \sinh \Delta\xi$; their product expands to $\sigma(\xi_*)^2(e^{\Delta\xi} - 2 + e^{-\Delta\xi}) = 2\sigma(\xi_*)^2(\cosh \Delta\xi - 1)$. □

Integrated slopes above and below any fixed point pack into hyperbolic-trig functions of rapidity displacement. The curve carries a Lorentzian signature in $(\xi, \ln \sigma)$ coordinates.

4 Reciprocal-strike symmetry

Proposition 2 (Mode as geometric center)**.** *Let ξ_C, ξ_P be rapidities with $\xi_C - \xi_m = -(\xi_P - \xi_m) =: \Delta \geq 0$. Then:*

- *(i) The slopes are reciprocals: $\sigma(\xi_C)\, \sigma(\xi_P) = 1$.*
- *(ii) The tangent angles satisfy $\tan \theta_C \, \tan \theta_P = \tan^2 \theta_m$: the mode is the geometric center of the curve in tangent-angle.*
- *(iii) The call premium at ξ_C equals the put premium at ξ_P, both equal to $e^{-\Delta}$, independently of w:*

$$P_C(\xi_C) \;=\; P_P(\xi_P) \;=\; e^{-\Delta}. \tag{14}$$

- *(iv) The same-side cross-premia are reciprocals: $P_C(\xi_C)\, P_C(\xi_P) = 1$, and $P_P(\xi_C)\, P_P(\xi_P) = 1$.*

Proof. (i) From (9), $\sigma(\xi_C)\, \sigma(\xi_P) = e^{(\xi_C + \xi_P) - 2\xi_m} = e^0 = 1$ since $\xi_C + \xi_P = 2\xi_m$. (ii) Exponentiate: $\tan \theta_C \tan \theta_P = e^{\xi_C + \xi_P} = e^{2\xi_m} = \tan^2 \theta_m$. (iii) $P_C(\xi_C) = e^{-(\xi_C - \xi_m)} = e^{-\Delta}$ and $P_P(\xi_P) = e^{\xi_P - \xi_m} = e^{-\Delta}$. (iv) $P_C(\xi_C) P_C(\xi_P) = e^{-(\xi_C - \xi_m) - (\xi_P - \xi_m)} = e^0 = 1$. □

Remark (The visible vs. invisible asymmetry)**.** Reciprocal-strike pairs carry identical option premia at every w. Pool tilt ($w \neq 1/2$) therefore *cannot* produce asymmetry in cross-side premia. What it does produce is asymmetry in the *polar angular distances* $\Delta\theta_C := \theta_C - \theta_m$ and $\Delta\theta_P := \theta_m - \theta_P$, which are unequal whenever $w \neq 1/2$ even though the corresponding premia agree. The asymmetry lives in geometry, not in price.

5 Trade dynamics: the warp

5.1 The warp rule

Definition (Curve warp)**.** A trade executed at point $(x_*, y_*) \in V(\cdot) = k_0$ with effective marginal price $P_{\text{eff}} := -dx/dy$ *(slippage-included secant)* deforms the pool state $(w_0, k_0) \to (w_1, k_1)$ by

$$w_1 := \frac{x_*}{x_* + P_{\text{eff}}\, y_*}, \tag{15}$$

$$k_1 := x_*^{w_1}\, y_*^{1-w_1}. \tag{16}$$

By construction, the new curve $V(\cdot) = k_1$ passes through (x_*, y_*) with marginal P_C on the new curve at the trade point equal to P_{eff}. Reserves do not move; the pricing function itself deforms.

5.2 Geometric reading

Proposition 3 (Warp is a rotation in log-log)**.** *Let* $(u_*, v_*) := (\ln x_*, \ln y_*)$*. The warp* (15)–(16)*:*

- *fixes the point* (u_*, v_*)*;*
- *rotates the line* $wu + (1-w)v = \ln k$ *about* (u_*, v_*)*, changing its slope from* $-w_0/(1-w_0)$ *to* $-w_1/(1-w_1)$*;*
- *slides the mode along the diagonal* $u = v$ *from* $(\ln k_0, \ln k_0)$ *to* $(\ln k_1, \ln k_1)$*.*

Proof. By (16), $V_1(x_*, y_*) = k_1$, so the trade point lies on the new curve. By (15), $-w_1/(1-w_1) \cdot y_*/x_* = -1/P_{\text{eff}}$, which is the new tangent slope at (x_*, y_*). The mode is the intersection with $u = v$, giving $(\ln k_j, \ln k_j)$. □

5.3 Trade impact

Theorem 2 (Mode-rapidity shift equals log marginal-price impact)**.** *With* P_0 *the pre-trade marginal price at the trade point,*

$$\Delta\xi_m \;:=\; \xi_m^{(1)} - \xi_m^{(0)} \;=\; \ln\frac{P_{\text{eff}}}{P_0}. \tag{17}$$

Proof. By (8), $\xi_m^{(j)} = \ln\big((1-w_j)/w_j\big)$. From (15), $(1-w_1)/w_1 = P_{\text{eff}}\, y_*/x_*$; from (4) at the trade point on the pre-trade curve, $(1-w_0)/w_0 = P_0\, y_*/x_*$. Take the ratio and log. □

Corollary 3 (Option-premium impact at the trade point)**.** *The trade point's call and put premia rescale by reciprocal factors:*

$$\frac{P_C^{(1)}(\xi_*)}{P_C^{(0)}(\xi_*)} \;=\; e^{\Delta\xi_m}, \qquad \frac{P_P^{(1)}(\xi_*)}{P_P^{(0)}(\xi_*)} \;=\; e^{-\Delta\xi_m}. \tag{18}$$

Proof. At fixed ξ_*, $P_C(\xi_*) = e^{\xi_m - \xi_*}$ scales as $e^{\Delta\xi_m}$; $P_P(\xi_*) = e^{\xi_* - \xi_m}$ scales as $e^{-\Delta\xi_m}$. □

Corollary 4 (Slippage)**.** *Define trader slippage as* $s := P_{\text{opt,eff}}/P_{\text{opt},0} - 1$*, with* P_{opt} *the side-specific option premium. Then*

$$s_{\text{call}} = e^{\Delta\xi_m} - 1, \qquad s_{\text{put}} = e^{-\Delta\xi_m} - 1. \tag{19}$$

A single scalar $\Delta\xi_m$ encodes the trade's full effect at the pivot. Slippage is $e^{\pm\Delta\xi_m} - 1$, with the sign set by the trade side.

5.4 Strike type-flip

Corollary 5 (Strike type-flip)**.** *Let ξ_{strike} be the rapidity of a strike on the book. The strike's classification as a call (if $\xi_{\text{strike}} > \xi_m$) or put (if $\xi_{\text{strike}} < \xi_m$) is determined by the side of ξ_m on which it lies. Hence a trade with $\Delta\xi_m \neq 0$ flips the type of every strike whose rapidity lies strictly between $\xi_m^{(0)}$ and $\xi_m^{(1)}$:*

$$\min(\xi_m^{(0)}, \xi_m^{(1)}) \; < \; \xi_{\text{strike}} \; < \; \max(\xi_m^{(0)}, \xi_m^{(1)}) \quad \Longrightarrow \quad \textit{strike type flips.}$$

The strike's intrinsic position (ξ_{strike}) does not move; only its classification relative to the moving mode reference does. Strikes outside the swept band $(\xi_m^{(0)}, \xi_m^{(1)})$ retain type and merely reprice per Corollary 3.

5.5 Visual summary

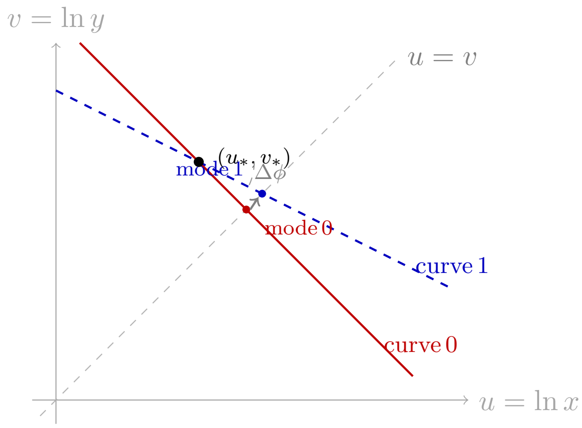

Figure: Warp in log-log (u, v) coordinates. The trade point (u_*, v_*) is fixed; the pricing line rotates through angle $\Delta\phi$ about it; the mode slides along the diagonal $u = v$ from mode 0 to mode 1. The mode-rapidity shift $\Delta\xi_m = \ln(P_{\text{eff}}/P_0)$ is the log of the price impact (Theorem 2). Strikes with ξ_{strike} between the two modes flip type.

6 Summary

Object	**Identification**
Pool curve	Straight line in $(\ln x, \ln y)$
Pool weight w	Line slope: $-w/(1-w)$
Pool scale k	Diagonal intersection: $(\ln k, \ln k)$
Rapidity ξ	$\ln\tan\theta = \ln y - \ln x$, intrinsic curve coord
Strike	Trade-invariant ξ_{strike} (or θ_{strike})
Mode rapidity ξ_m	$\ln\big((1-w)/w\big)$; moves under trade
Marginal slope	$\sigma(\xi) = e^{\xi - \xi_m}$
Call premium	$P_C(\xi) = e^{\xi_m - \xi}$
Put premium	$P_P(\xi) = e^{\xi - \xi_m}$
Premium duality	$P_C \cdot P_P \equiv 1$; both $= 1$ at mode (ATM)
Conservation (Thm 1)	$\sigma(\xi + \Delta\xi) \cdot \sigma(\xi - \Delta\xi) = \sigma(\xi)^2$
Reciprocal pair (Prop 2)	$\xi_C + \xi_P = 2\xi_m \Rightarrow P_C(\xi_C) = P_P(\xi_P)$
Warp action	Rotation of line about (u_*, v_*)
Mode shift (Thm 2)	$\Delta\xi_m = \ln(P_{\text{eff}}/P_0)$
Premium impact	P_C scales by $e^{\Delta\xi_m}$; P_P by $e^{-\Delta\xi_m}$
Slippage (call / put)	$e^{\Delta\xi_m} - 1$ / $e^{-\Delta\xi_m} - 1$
Strike type-flip (Cor 5)	$\xi_{\text{strike}} \in (\xi_m^{(0)}, \xi_m^{(1)}) \Rightarrow$ type flips

The AMM is a one-dimensional rapidity object. Strikes are fixed points on this line; the mode is a moving cursor; trades shift the cursor by an amount equal to the log of the price impact. Premia and slopes are evaluations at $\xi_{\text{strike}} - \xi_m$; the entire pool's response to a trade is one scalar.